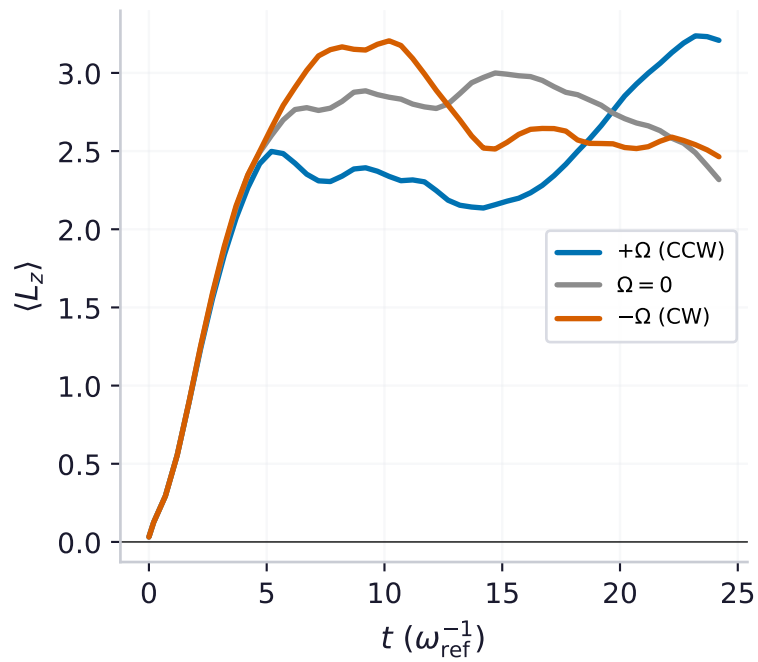
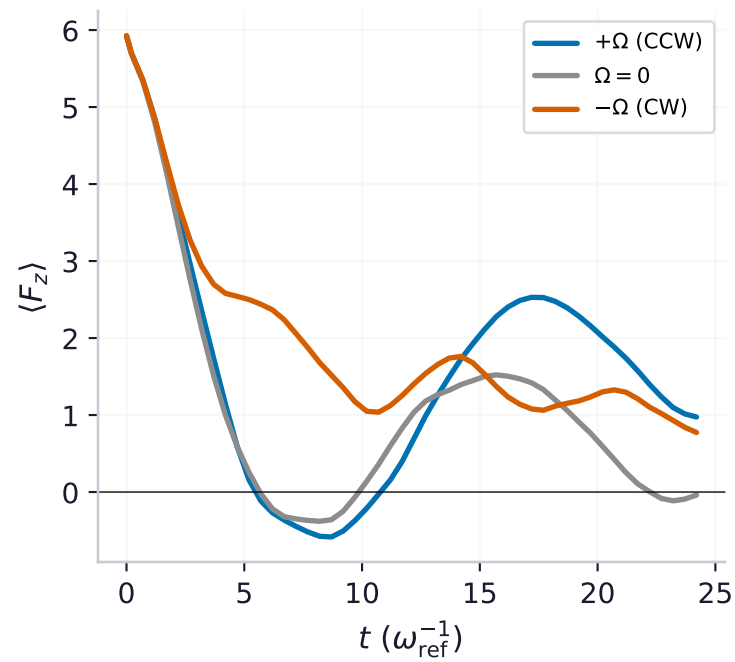


Rotating magnetic field: direction-dependent vortex + Barnett spin response (^{151}Eu $F=6$, unitary)

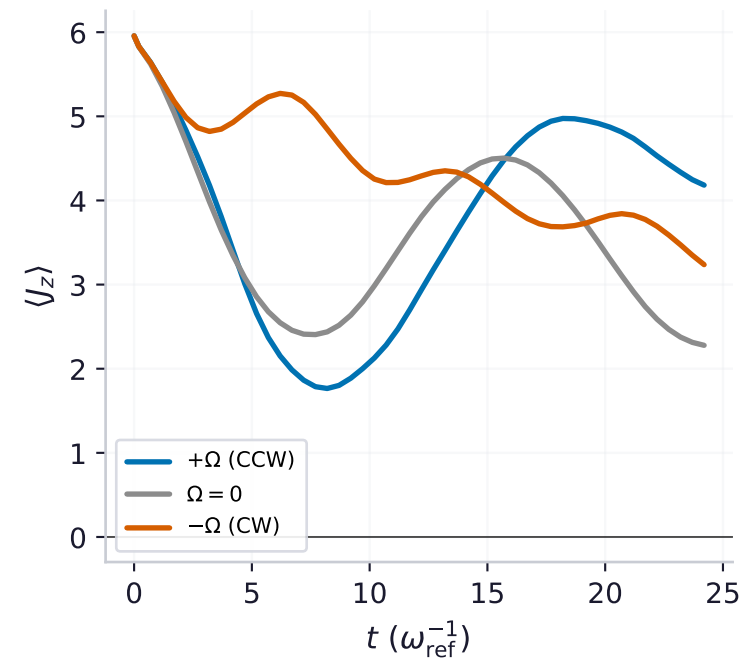
(a) orbital $\langle L_z \rangle$ = vortex response



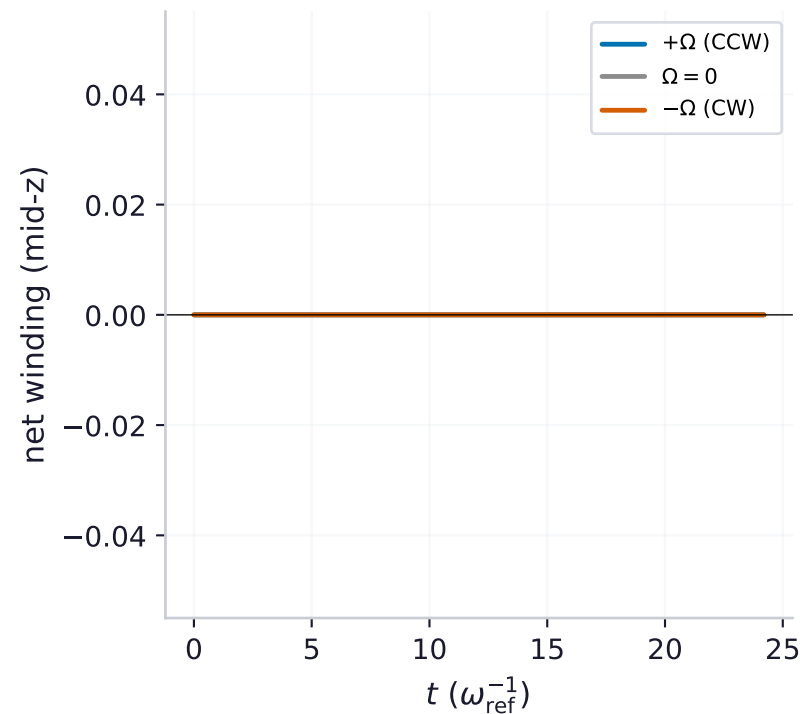
(b) Barnett spin $\langle F_z \rangle$



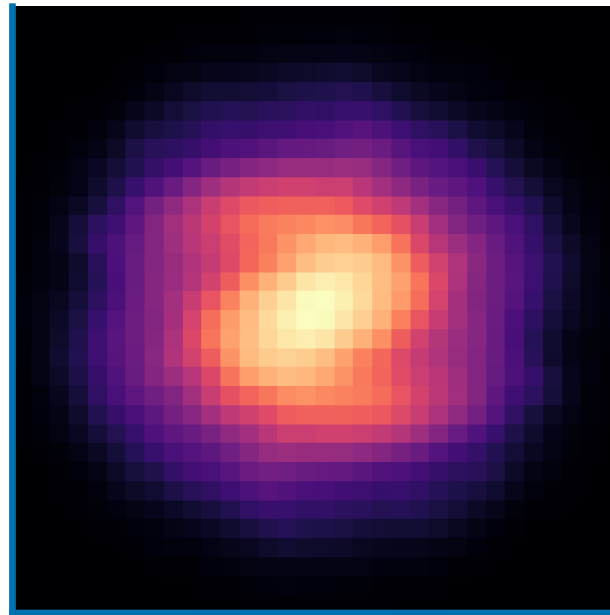
(c) $\langle J_z \rangle$ (field pumps AM)



(d) vortex chirality vs rotation



$+\Omega$ (CCW): total density (late)



$-\Omega$ (CW): total density (late)

